

American Put Option Pricing, SOC'26

Report on HDFC Bank, option pricing

I have selected HDFC Bank options for this assignment. Below are the needed parameters.

Option Chain (Equity Derivatives)

Futures contracts

View Options Contracts for:

Select Symbol

Expiry Date

Strike Price

Select

OR

HDFCBANK

30-Jun-2026

OR

Select

Underlying Symbol : HDFCBANK 739.70

As on 09-Jun-2026 15:30:00 IST

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Best View

Download (.csv)

CALLS													PUTS												
	OI	CHNG IN OI	VOLUME	IV	LTP	CHNG	BID QTY	BID	ASK	ASK QTY	STRIKE	BID QTY	BID	ASK	ASK QTY	CHNG	LTP	IV	VOLUME	CHNG IN OI	OI				
✓	1,092	102	4,396	11.05	20.55	0.90	550	20.50	20.80	1,100	725.00	550	12.65	12.80	1,100	-1.95	12.80	29.90	4,147	211	2,192	✓			
✓	5,232	864	23,929	13.86	18.20	0.95	1,100	17.95	18.20	3,300	730.00	550	15.10	15.30	5,500	-1.95	15.25	30.65	9,718	584	7,139	✓			
✓	2,446	418	12,858	15.52	15.95	1.00	1,650	15.75	15.95	4,400	735.00	550	17.80	18.00	1,650	-2.25	17.75	31.16	4,275	195	2,313	✓			
✓	12,489	2,495	35,886	16.89	14.00	0.95	1,650	13.85	14.00	75,900	740.00	1,650	20.80	21.05	1,100	-1.95	21.05	32.54	7,104	664	7,932	✓			
✓	3,530	92	9,318	17.87	12.15	0.85	1,650	12.10	12.20	550	745.00	1,100	24.00	24.30	1,100	-1.85	24.20	33.44	1,614	-44	1,676	✓			
✓	21,775	5,086	31,182	18.91	10.65	0.85	2,750	10.65	10.80	32,450	750.00	1,100	27.50	27.80	1,650	-1.85	27.80	34.74	2,978	556	13,285	✓			
✓	5,170	681	6,758	19.72	9.25	0.70	6,600	9.25	9.35	550	755.00	550	30.90	31.60	550	-1.20	32.00	36.69	498	-121	1,681	✓			

Spot (S) = 739.7 INR

Strike Price (K)=740

T = 21/365.25 = 0.057 (approx.)

Estimating Volatility

To estimate historical volatility, daily closing prices of HDFC Bank Ltd. from 09 March 2026 to 09 June 2026 were obtained from NSE historical market data. Daily log returns were computed using:

$$r_t = \ln(P_t/P_{t-1})$$

where (P_t) is the closing price on day (t).

The standard deviation of the most recent 30 daily log returns was then calculated and annualized using:

$$\sigma = s \cdot \sqrt{252} = 21.76\%$$

where (s) is the sample standard deviation of daily log returns and 252 represents the approximate number of trading days in a year.

Using this, the annualized historical volatility (σ) = 21.76%

The selected option for analysis was the 30 June 2026 expiry contract with strike price ₹740.

According to the NSE option chain, the implied volatility (IV) for the selected strike was approximately 16.19% for the call option and 33.27% for the put option. (shown in screenshot of NSE)

Measure	Volatility
Historical Volatility	21.76%
Call Implied Volatility	16.19%
Put Implied Volatility	33.27%

The Gap :- The call option's implied volatility is lower than the historical volatility (-5.57% difference), suggesting that the market expects future price fluctuations to be lower than those observed during the recent period. In contrast, the put option's implied volatility is substantially higher than the historical volatility (11.51% difference), indicating stronger demand for downside protection and the presence of volatility skew. This difference illustrates a limitation of the Black–Scholes model, which assumes a single constant volatility parameter across all strikes and option types.

3.

I have decided to use RBI repo rate to calculate the compounded r value as it is the industry standard for educational purposes and provides some sense of uniformity throughout examples in India.

At the time of analysis, the Policy Repo Rate was 5.25% per annum.

Policy Repo Rate (R) = 5.25%

$$r = \ln(1 + R)$$

here R is RBI repo rate.

$$r = \ln(1.0525)$$

$$r = 0.05117$$

$r = 5.12\%$ per annum is our risk free compounded rate.

4.

$$\begin{aligned}
 d_1 &= \frac{\ln(S/K) + (r + \sigma^2/2) \times T}{\sigma \times \sqrt{T}} \\
 &= \frac{\ln\left(\frac{739.7}{740}\right) + (0.05117 + 0.2176^2/2) \times 0.0575}{0.2176 \times \sqrt{0.0575}} \\
 &= \frac{0.003899}{0.05218} = 0.07473 \\
 d_2 &= d_1 - \sigma \times \sqrt{T} \\
 &= 0.07473 - 0.05218 \\
 &\approx 0.02254
 \end{aligned}$$

The values of $N(d_1)$ and $N(d_2)$ were obtained using the cumulative standard normal distribution. For $d_1 = 0.0747$ and $d_2 = 0.0225$, the corresponding cumulative probabilities are:

$$N(d_1) = N(0.0747) = 0.5298$$

$$N(d_2) = N(0.0225) = 0.5090$$

$$C = S \times N(d_1) - K \times e^{-rt} \times N(d_2)$$

$$= 739.7 \times 0.5298 - 740 \times e^{-0.05117 \times 0.057} \times 0.5090$$

$$= 391.90 - 375.56$$

$$= ₹16.34$$

By Put-Call Parity,

$$C - P = S - K e^{-rt}$$

$$P = ₹14.80$$

5.

The Black–Scholes model estimated a call value of ₹16.34 and a put value of ₹14.46, compared with market prices of ₹14.00 and ₹21.05, respectively. The call option traded slightly below its theoretical value, while the put option traded significantly above it. This suggests that the market is relatively more bearish and risk-averse than the Black–Scholes model implies, as investors are willing to pay a premium for downside protection. The large difference between call IV (16.89%) and put IV (32.54%) indicates the presence of volatility smile/skew effects, showing that volatility is not constant across options. This deviation shows a key limitation of the Black–Scholes model, which assumes a single constant volatility parameter.

6.

Greek	Value	Interpretation
Delta	0.5298	The call price increases by approximately ₹0.53 for every ₹1 increase in HDFCBANK's stock price.
Gamma	0.0103	Delta changes by approximately 0.0103 for every ₹1 change in the stock price.
Vega	70.59	For a 100 percentage-point increase in volatility, the option value changes by ₹70.59. Therefore, a 1 percentage-point increase in volatility increases the option value by approximately ₹0.71.

Theta	-152.70 per year	Time decay reduces option value as expiry approaches. Equivalent to approximately -₹0.42 per day ((-152.70/365)).
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- For 1% increase in S:
Delta tells us the change which would be 1% of stock price * value of delta.

$$\Delta C \approx ₹3.92$$

$$\text{Call value} = 16.34 + 3.92 = ₹20.26$$

- For 1% increase in volatility:
Vega tells us this (by definition),
 $\Delta C \approx \text{Vega} \times 0.01$
 $\Delta C \approx 70.59 \times 0.01$
 $\Delta C \approx ₹0.71$
Call value = 17.05 INR

7.

HDFCBANK stock options are American-style options, meaning they may be exercised before expiry. However, for the selected 740 strike, the early-exercise premium is likely to be small. The option is approximately at-the-money ($S = ₹739.70$, $K = ₹740$) and still has about 21 days remaining until expiry, so it retains significant time value. Early exercise is generally optimal for put options that are deep in-the-money, close to expiry, and have little remaining time value. Therefore, although an American put should theoretically be worth at least as much as a comparable European put, the difference in value for this contract is expected to be small.

8.

While calculating the values of greeks for this assignment, my intuition on the amount gained during volatility changes was broken. I had expected large trends based on volatility but the difference caused by it wasn't as large as I had in mind, this helps me see that the market is not always swayed by micro trends even in highly volatile stocks.

Sources :- ChatGPT (for last minute editing)

[video on greeks](#)

[Investopedia](#)